MULTILINGUAL MULTIMODAL TOURISM AGENT FOR SRI LANKA USING KNOWLEDGE GRAPH REASONING

Project Id: 25-26J-119

Project Proposal Report

Fernando T K

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

MULTILINGUAL MULTIMODAL TOURISM AGENT FOR SRI LANKA USING KNOWLEDGE GRAPH REASONING

Project Id: 25-26J-119

Project Proposal Report

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Fernando T K	IT22059536	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision

Signature of the supervisor: Date: 29/08/2025

ABSTRACT

Sri Lanka's tourism sector is increasingly turning to intelligent digital assistants that can provide personalized, sustainable, and proactive guidance to travelers. This project proposes the development of a Multilingual and Multimodal Tourism Agent for Sri Lanka Using Knowledge Graph Reasoning, structured into four interconnected components. The first component delivers a multilingual and multimodal language service, enabling Sinhala–English code-mixed speech recognition, text-to-speech with emotional tone, and Sinhala OCR-to-Singlish transliteration. The second component develops the Sri Lankan Tourism Knowledge Graph and AI Reasoning Layer, integrating data from transport, accommodation, pricing, events, and weather into a semantically structured knowledge base with explainable reasoning capabilities. The fourth component designs a multi-agent orchestration framework, enabling coordination across domain-specific agents (transport, accommodation, visa, events) using comparative protocols such as MCP, A2A, and blackboard architectures. Together, these modules establish a cohesive and adaptive tourism ecosystem.

Within this ecosystem, the third component—Proactive Tourism Recommendation and Sustainability Service—forms the decision-making core for traveler guidance. This service integrates predictive analytics to forecast demand, crowd levels, and pricing trends, re-ranks recommendations using multi-objective optimization that balances personalization, cost, and sustainability, and provides proactive "micro-moment" interventions to deliver timely, context-aware suggestions. Automated itinerary planning is supported through advanced formulations of the Tourist Trip Design Problem (TTDP), ensuring optimized multi-day plans under real-world constraints. The methodology employs a hybrid architecture of predictive models, a knowledge graph for semantic reasoning, and emerging neuro-symbolic approaches that unify reasoning with learning.

Anticipated outcomes include improved recommendation relevance, higher itinerary satisfaction, and measurable reductions in overtourism pressure through sustainability-aware re-ranking. By integrating multilingual interaction, real-time predictive analytics, and knowledge graph reasoning, this research contributes a pioneering framework for intelligent destination management in Sri Lanka, with potential for commercialization as a smart tourism concierge platform.

Keywords — Tourism Knowledge Graph; Multilingual and Multimodal Interaction; Sustainable Tourism; Proactive Recommendation Systems; Multi-Agent Orchestration

Table of Contents

DECLARATION i
ABSTRACT ii
LIST OF FIGURES iv
LIST OF TABLES v
LIST OF ABBREVIATIONS vi
1.INTRODUCTION 1
1.1 Background and Literature Review 2
1.2 Research Gap 5
1.3 Research Problem 8
2. OBJECTIVES 10
2.1 Main Objective 10
2.2 Specific Objectives 10
3. METHODOLOGY 13
4.PROJECT REQUIREMENTS 20
4.1 Functional Requirements 20
4.2 Non-Functional Requirements 21
4.3 System Requirements 22
4.4 User Requirements 22
5. GANTT CHART 23
5.1 Work Breakdown Structure 23
6.PERSONAL AND FACILITIES 24
7. BUDGET AND BUDGET JUSTIFICATION 24
7. REFERENCES 27

LIST OF FIGURES

Figure 3.1:Overall System Diagram 13
Figure 3.2: Component Diagram 17
Figure 5.1:Gantt Chart 23
Figure 5.2: Work breakdown structure 23

LIST OF TABLES

Table 1.1:Existing Research Comparison 7
Table 7.1Budget Table 25

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
A2A	Agent-to-Agent (direct communication between agents)
CER	Character Error Rate
GA	Genetic Algorithm
IoT	Internet of Things
KG	Knowledge Graph
KGCN	Knowledge Graph Convolutional Network
LSTM	Long Short-Term Memory (neural network model)
ML	Machine Learning
MOS	Mean Opinion Score (subjective quality rating, e.g., TTS)
MOPSO	Multi-Objective Particle Swarm Optimization
MCP	Master Control Protocol (coordination strategy in agent orchestration)
NFR	Non-Functional Requirement
NSGA-II	Non-Dominated Sorting Genetic Algorithm II (multi-objective optimization)
NLP	Natural Language Processing
OCR	Optical Character Recognition
p95	95th Percentile (performance measurement)
QoS	Quality of Service
RS	Recommender System
SPARQL	SPARQL Protocol and RDF Query Language
STT	Speech-to-Text
TTS	Text-to-Speech
TTDP	Tourist Trip Design Problem

1.INTRODUCTION

Sri Lanka's tourism industry is poised to benefit from next-generation intelligent systems that combine multilingual interaction, multimodal understanding, predictive analytics, and knowledge graph reasoning. Tourists today demand more than static guides or reactive recommendations—they expect context-aware digital assistants that can proactively adapt to real-time conditions such as weather, prices, and crowd density, while also ensuring cultural sensitivity and sustainability [2],[18]. Globally, intelligent tourism platforms are beginning to integrate knowledge graph reasoning [6], conversational AI [10], and proactive recommendation techniques [2]; however, most focus on European and Chinese contexts, with limited support for Sinhala–English multimodality and localized sustainability needs. To address this gap, we propose a Multilingual and Multimodal Tourism Agent for Sri Lanka Using Knowledge Graph Reasoning, a four-component framework designed to deliver an adaptive and sustainable digital tourism experience.

The project is divided into four tightly connected components. The first component provides a Multilingual and Multimodal Language Service, enabling Sinhala–English code-mixed speech-to-text, text-to-speech with emotion rendering, Sinhala OCR for signboards, and transliteration into Singlish. The second component develops a Sri Lankan Tourism Knowledge Graph with an Agent Interface Layer, aggregating transport, accommodation, events, pricing, and weather data into a semantically structured ontology, supporting SPARQL queries, reasoning, and explainability [1]. The fourth component implements Multi-Agent Orchestration and Contextual Coordination, enabling integration of domain agents (e.g., hotels, transport, visas, attractions) through different coordination strategies such as MCP, direct agent-to-agent (A2A), and blackboard architectures. Together, these components provide the infrastructure for a digital tourism ecosystem that is multilingual, adaptive, and scalable.

Within this ecosystem, the third component—Proactive Tourism Recommendation and Sustainability Service—functions as the decision-making engine. Its novelty lies in

integrating predictive analytics to forecast demand, crowd density, and pricing [2],[17], while balancing personalization, budget constraints, and sustainability through multi-objective optimization [18][9]. Unlike conventional recommender systems that rely on user-initiated queries, this service introduces proactive "micro-moment" interventions that deliver timely, context-aware suggestions during a trip [3][4]. Furthermore, it supports automated itinerary planning by solving the Tourist Trip Design Problem (TTDP), producing optimized daily schedules under predictive conditions [16]. To enhance reasoning and adaptivity, the system leverages the knowledge graph [1] and considers emerging neuro-symbolic approaches that unify statistical learning with symbolic reasoning for explainability and robustness [19].

In summary, this project integrates multilingual interaction, knowledge graph reasoning, proactive recommendation, and sustainable tourism management into a unified system tailored for Sri Lanka. To achieve this, the project requires mastering techniques in multilingual speech and text processing, semantic web technologies, predictive modeling, and multi-agent orchestration. The specific contribution of this research lies in my Component, which delivers a predictive, sustainable, and proactive recommendation service that not only enhances traveler satisfaction but also addresses pressing challenges such as overtourism and environmental impact.

1.1 Background and Literature Review

Tourism recommender systems (RS) have evolved significantly, moving from static, popularity-based suggestions to intelligent, context-aware engines. However, personalization in tourism is highly challenging because of dynamic conditions such as weather, visitor flows, pricing, and sustainability concerns. Component of this project—the Proactive Tourism Recommendation and Sustainability Service—is designed to address these challenges by integrating predictive forecasting, sustainability-aware re-ranking, and proactive micro-moment interventions.

1. **Predictive Analytics in Tourism RS**

A distinguishing feature of modern systems is their ability to predict future travel conditions. Li and Zheng (2025) demonstrated that by forecasting risks such as bad weather or overcrowding, proactive tourism recommenders can help travelers avoid disruptions [2]. Similarly, R2Tour (2023) introduced a real-time data-driven framework that continuously ingests live streams to train tourism patterns and deliver foresight-driven recommendations [17]. These approaches highlight the need for predictive personalization that adapts recommendations before conditions change.

2. **Dynamic Pricing and Forecasting**

Dynamic pricing is a key signal for demand prediction in tourism. Guizzardi et al. (2021) showed that fluctuations in hotel and transport prices are strong indicators of tourist arrivals [5]. By incorporating predictive models for price movements, recommender systems can alert travelers to book at optimal times or suggest alternatives when costs are forecasted to rise. This component builds on these findings by integrating crowd, weather, and price forecasting models (e.g., LSTM, Prophet) into its recommendation pipeline.

3. **Sustainability and Overtourism-Aware Recommendations**

Sustainability has become a central theme in tourism recommender research. Nguyen et al. (2024) introduced OurSCARA, which encouraged eco-friendly attractions and community-based experiences [18]. Reinforcement learning–based approaches have also been applied to rerank attraction sequences in order to avoid over-congestion [9]. Our service extends these ideas by embedding a Sustainability Re-Ranking Module that balances personalization with overtourism mitigation, ensuring recommendations support both traveler satisfaction and environmental stewardship.

Proactive, Context-Aware, and Micro-Moment Engagement

Traditional systems rely on “pull-based” interaction where users explicitly request suggestions. Proactive systems instead push recommendations in micro-moments, short intent-rich intervals when tourists need immediate guidance [3][4]. By combining contextual triggers (time, location, weather) with predictive models, our service aims to function as a virtual concierge that delivers timely, value-added interventions, such as suggesting indoor activities before a forecasted storm.

4. **Automated Itinerary Planning (TTDP)**

The Tourist Trip Design Problem (TTDP) frames itinerary planning as an optimization challenge that balances user preferences, time, and constraints. Zhang et al. (2022) modeled TTDP with time windows, producing multi-day routes under real-world constraints [16]. Our system advances this by embedding predictive analytics—adjusting itineraries dynamically when forecasts indicate high crowd levels, adverse weather, or sudden price surges.

5. **Knowledge Graph Reasoning and Neuro-Symbolic Trends**

Tourism knowledge graphs improve explainability and support complex queries, such as "find eco-friendly cultural attractions near Kandy with moderate crowds." Recent work highlights the potential of KG-based tourism architectures [1], enabling evidence-backed recommendations. In parallel, neuro-symbolic approaches are emerging as a way to unify statistical learning with symbolic reasoning, offering greater adaptability and transparency in intelligent recommendation engines [19]. Our system leverages both directions to enhance explainability and robustness.

Summary of Insights

The literature confirms strong progress in predictive forecasting [2][17], dynamic pricing [5], sustainability-aware ranking [18][9], micro-moment proactivity [3][4], and itinerary optimization [16]. However, no single system integrates these into a unified recommendation framework that simultaneously forecasts, optimizes, and re-ranks itineraries for personalization and sustainability in a Sri Lankan context. This gap motivates the design of Component: a predictive, sustainable, and proactive tourism recommender that balances traveler experience with sustainable destination management.

1.2 Research Gap

Despite rapid advances in tourism recommender systems, a clear research gap remains in delivering predictive, sustainable, and proactive recommendations tailored to real-world conditions in Sri Lanka. Existing works address isolated aspects-prediction, sustainability, or itinerary optimization but fail to unify them into a single decision-making framework.

1. **Limited Use of Predictive Analytics in Personal Recommenders**

While forecasting techniques such as demand prediction and crowd modeling exist in tourism operations research, most recommender systems do not embed them into live traveler-facing platforms. Li and Zheng (2025) demonstrated how forecasting could preemptively optimize recommendations [2], and R2Tour (2023) showed the value of real-time data ingestion for training [17]. However, no deployed system integrates short-term forecasting of crowds, weather, and prices into personalized recommendation delivery.

2. **Lack of Unified Multi-Objective Optimization**

Tourism recommenders often prioritize either personalization or a single factor such as location or cost. Few attempt to jointly optimize for traveler satisfaction, budget efficiency, and sustainability. Systems like OurSCARA [18] encourage eco-friendly choices, and reinforcement learning–based approaches mitigate overtourism [9], but none simultaneously balance personalization, affordability, and ecological impact through multi-objective optimization.

3. **Underutilization of Proactive Micro-Moment Engagement**

Micro-moments—short, intent-rich intervals requiring immediate action—are underexplored in deployed systems. Current platforms remain largely reactive, waiting for explicit user queries. Although research has highlighted the value of proactive interventions [3][4], there is little evidence of real-time, context-driven recommendation delivery, particularly in Sri Lanka's tourism context.

4. **Incomplete Integration of Knowledge Graph Reasoning**

Knowledge graphs enhance semantic reasoning and explainability, yet their adoption in real-world tourism recommenders remains limited. Recent frameworks show promise in KG-driven reasoning [1], but few combine KG queries with predictive forecasting to deliver transparent, evidence-based recommendations.

5. Lack of Explainable Neuro-Symbolic Approaches

Neuro-symbolic AI offers a promising direction for blending statistical learning with symbolic reasoning, improving robustness and interpretability [19]. However, its application in tourism recommendation is unexplored, leaving a gap in explainability and user trust.

6. Limited Integration of Predictive Context into Itinerary Planning

Itinerary planning models (TTDP) typically optimize routes based on static preferences or heuristics [16]. They rarely adapt dynamically to forecasted conditions such as sudden weather changes, visitor surges, or price fluctuations. As a result, generated itineraries may be feasible in design but unrealistic in practice.

Feature / Dimension	**A**	**B**	**C**	**D**	**Our Proposed System**
Personalized Recommendation	✓	✓	✓	✓	✓
Predictive Analytics	Partial	✗	✗	✗	✓
Sustainability Optimization	✗	✓	✗	✗	✓
(Micro-Moments)	✗	✗	✗	✗	✓
Knowledge Graph Reasoning	✗	✗	✗	✓	✓
Explainability	✗		✗	Partial	✓

Dynamic Itinerary Adaptation	✗	✗	✓	✗	✓

Table 1.1:Existing Research Comparison

A — Real-time context-aware recommendation system for tourism [17] .

B — Knowledge-Graph–Based Tourism Recommender Systems: KG-driven personalization and semantic reasoning.[1]
C — Predictive Analytics for Dynamic Pricing in Travel Bookings: Demand/price forecasting models for airlines/hotels.[15]
D — Tourism Recommendation System Based on Semantic Clustering and Sentiment Analysis: Context-aware TRS using semantic clustering of reviews + sentiment.[14]

Our system unifies short-term forecasting (crowd/weather/price), KG reasoning, and sentiment-derived preferences into one traveler-facing engine. Unlike prior single-focus systems, it optimizes personalization, budget, and sustainability together, triggers proactive micro-moment interventions, and explains choices via neuro-symbolic logic—while dynamically adapting itineraries for Sri Lanka.

Summary of the Gap

Existing studies have explored predictive analytics [2][17], sustainability-aware optimization [18][9], micro-moment engagement [3][4], knowledge graph reasoning [1], and neuro-symbolic explainability [19] as separate strands. However, no tourism recommender system integrates all of these dimensions into a single predictive, sustainable, and proactive framework. This research addresses the gap by designing a decision-making engine that unifies these elements within a Sri Lankan tourism knowledge ecosystem, directly contributing to more relevant, explainable, and sustainable travel experiences.

1.3 Research Problem

Given the identified gap, the central research problem can be formulated as:
How can we design and implement a tourism recommendation engine that proactively delivers personalized suggestions and optimized itineraries to travelers by predicting future conditions and optimizing for multiple objectives (user preferences, budget, and sustainability), in order to enhance traveler experience and promote sustainable tourism in Sri Lanka?

This problem encapsulates several interrelated challenges:

1. **Predictive Personalization**

In what ways can predictive models for weather, crowd density, and pricing be integrated into the recommendation process so that suggestions are not only context-aware but also foresight-driven? Recent systems such as R2Tour demonstrate the value of real-time forecasting for adaptive tourism recommendations [17], but no system has yet delivered these capabilities in live, traveler-facing platforms.

2. **Multi-Objective Trade-offs**

How can the trade-off between a traveler's immediate satisfaction (e.g., visiting a popular landmark) and long-term sustainability (e.g., mitigating overtourism and reducing environmental impact) be modeled and optimized? Prior work has introduced sustainability-aware re-ranking [18][9], but combining this with personalization and budget optimization remains an open challenge.

3. **Automated Itinerary Planning**

How can algorithms generate complete, multi-day itineraries that account for user constraints (trip duration, budget, preferences) while also adapting to predictive contexts such as future weather or crowd fluctuations? Existing Tourist Trip Design Problem

(TTDP) approaches optimize static routes [16], but the challenge lies in integrating predictive analytics to keep itineraries realistic under changing conditions.

4. **Proactive Engagement**

How can micro-moment detection be leveraged to deliver timely recommendations without explicit user queries, ensuring that interventions are contextually valuable rather than disruptive? This extends prior work on micro-moment–based frameworks for proactive interaction [3][4], but has not yet been realized in Sri Lanka's tourism domain.

Addressing these challenges collectively will enable the development of a foresight-driven, sustainable, and proactive tourism recommender tailored to Sri Lanka's multilingual and culturally sensitive context, offering both academic novelty and practical impact.

2. OBJECTIVES

2.1 Main Objective

Design and develop a Predictive, Sustainable, and Proactive Tourism Recommendation Engine that enhances traveler decision-making in Sri Lanka by providing personalized, context-aware suggestions and optimized itineraries, tailored to user satisfaction, budget constraints, and sustainability considerations.

2.2 Specific Objectives

1. Develop Real-Time Data Integration and Forecasting Models

- Specific: Implement modules to continuously collect real-time tourism-related data (weather updates, crowd density, attraction footfall, prices) and train predictive models (e.g., LSTM or Prophet for time series) to forecast short-term future conditions. Approaches like R2Tour demonstrate the value of real-time forecasting in tourism RS [17].
- Measurable: Achieve high accuracy in next-day visitor count predictions for major attractions.
- Achievable: Use publicly available APIs (e.g., meteorological data, Google Popular Times) and proven ML algorithms.
- Realistic: The team has data science expertise and will use existing libraries for forecasting.
- Time-bound: Data pipeline and initial forecasting model to be completed by Month 3.

2. Implement a Multi-Objective Recommendation Algorithm

- Specific: Design an optimization algorithm (e.g., NSGA-II or MOPSO) combining multiple criteria: (a) user preference fit, (b) cost-efficiency, and (c) sustainability index. Prior work highlights the need for balancing personalization with sustainability in recommendations [18][9].
- Measurable: Produce Pareto-optimal recommendation sets; most of the recommendations should stay within budget and reduce predicted overcrowding by compared to baseline.
- Achievable: Adapt known multi-criteria RS methods with custom factors.
- Realistic: Computation feasible server-side with optimization libraries.
- Time-bound: Prototype running by Month 5.

3. Incorporate Overtourism Mitigation and Sustainability Re-Ranking

- Specific: Develop a re-ranking module to adjust order of recommendations using sustainability scores (e.g., crowd level, ecological sensitivity). Similar sustainability-aware frameworks have proven effective in rebalancing flows [18][9].
- Measurable: High % of itineraries should include one "off-peak or alternative" suggestion; improvement in visitor distribution evenness.
- Achievable: Use crowd forecasts and carrying capacity approximations.
- Time-bound: Tested by Month 6.

4. Enable Automated Itinerary Planning

- Specific: Implement an itinerary planner that generates multi-day travel plans optimized for user-defined trip duration, budget, and preferences, while incorporating predictive context (weather, crowds, pricing). Solve the Tourist Trip Design Problem (TTDP) using metaheuristics such as genetic algorithms and orienteering formulations [16].

- Measurable: High percentage of generated itineraries should meet all hard constraints (time, budget, opening hours). Pilot evaluation should show improvement in traveler satisfaction with itineraries compared to static plans.
- Achievable: Leverage existing optimization frameworks and extend them with predictive features.
- Realistic: Datasets and solvers are available; focus is integration.
- Time-bound: Initial itinerary planner by Month 6; refined version with predictive adaptation by Month 9.

5. Enable Proactive Micro-Moment Recommendation Delivery

- Specific: Context recognition module to detect user state (location, time, activity) and trigger relevant push recommendations. This extends prior research on micro-moment tourism engagement [3][4].
- Measurable: Identify ≥5 distinct micro-moment contexts; aim for ≥50% user acceptance rate of proactive suggestions.
- Achievable: Use smartphone sensors and lightweight classifiers.
- Time-bound: Detection logic by Month 7; integrated delivery by Month 8.

7. User Interface and Explainability

- Specific: Design a user-facing interface (mobile or chatbot-integrated) with recommendation rationales (e.g., "Recommended because it matches your hiking interest and is less crowded this afternoon [9]"). Incorporate KG-driven explanations [1] and explore neuro-symbolic explainability for transparency [19].
- Measurable: explanation templates; user study participants find explanations clear/helpful.
- Achievable: Template-based generation from stored reason codes.
- Time-bound: Prototype UI and explanations by Month 10.

3. METHODOLOGY

The proposed system will be implemented through a modular, data-driven methodology. The approach combines software engineering for system integration with research and experimentation for algorithm development. This section provides an overview of the system architecture, data sources, algorithms, and development plan. A high-level system diagram (Figure 3.1) illustrates how the components interact to achieve the project objectives.

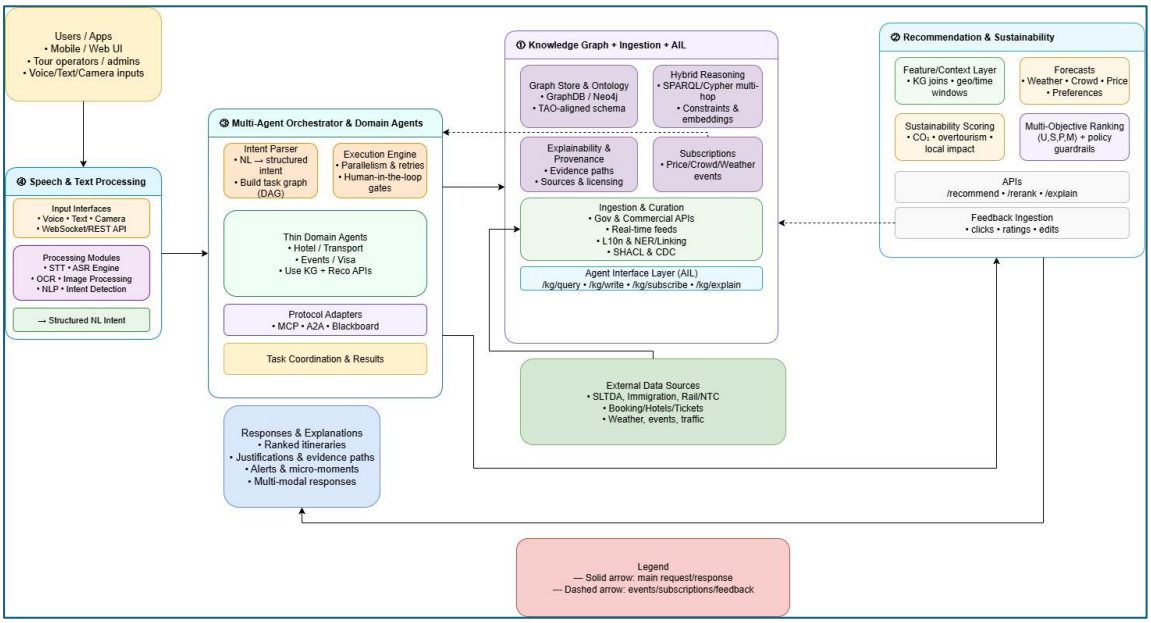

Figure 3.1:Overall System Diagram

Figure 3.1: High-level System Architecture and Data Flow for the Predictive, Sustainable, and Proactive Tourism Recommendation Engine. The system ingests real-time data from various sources into the Data Ingestion Layer, processes them with predictive analytics, and generates recommendations through a multi-objective optimization core. Sustainability assessment (overtourism mitigation), itinerary planning, and micro-moment context detection influence recommendation generation. A knowledge graph and user

profile feed into personalization and explanation modules. The final recommendations are delivered to the user interface, with continuous feedback loops for learning.

Overall System Description

The engine is designed as a service-oriented architecture with seven main layers or components working in sequence:

1. Real-Time Data Ingestion Layer

- This layer continuously gathers data streams relevant to tourism recommendations. Sub-components include:
- Weather Data Collector – pulling forecasts from APIs like OpenWeatherMap.
- Crowd Density Monitor – ingesting live crowd estimates from sources like Google Places or telecom data.
- Dynamic Pricing Tracker – monitoring hotel, flight, and attraction prices.
- User Context Analyzer – tracking user's location, time, and activity.
- The ingestion layer standardizes and stores these inputs in a time-series database (e.g., InfluxDB) and context store. For instance, if a storm is forecast for the evening, this information is immediately available to the recommendation and itinerary modules [2]. Real-time ingestion is a key enabler of foresight-driven recommendation systems such as R2Tour [17].

2. Predictive Analytics Engine

- This component transforms raw data into future context, producing short-term forecasts for weather, crowd density, and prices. Models include:
- Weather Prediction Model – tourism-specific forecasts.
- Crowd Forecasting Model – predicts attraction footfall using LSTM/ARIMA [9].
- Price Prediction Module – forecasts dynamic pricing trends [15].

- User Behavior Predictor – adapts recommendations based on prior interactions and feedback.
- These predictive signals feed into optimization, itinerary planning, and re-ranking components, enabling foresight-driven personalization [17].

3. Multi-Objective Optimization Core

- The heart of the decision-making engine formalizes recommendation as an optimization problem:
- Weights are tuned to balance satisfaction, cost, and sustainability. Algorithms such as NSGA-II or MOPSO will generate Pareto-optimal candidate recommendations. Prior work has shown the importance of balancing personalization with sustainability in RS design [18][9]. The system dynamically manages trade-offs (e.g., a crowded but iconic site vs. a serene alternative).

4. Automated Itinerary Planning Module

- Unlike conventional recommenders, our system includes a dedicated Itinerary Planner that generates multi-day tour sequences optimized under predictive conditions.
- Approach: Formulate the Tourist Trip Design Problem (TTDP) with constraints like trip duration, opening hours, travel time, and budget.
- Techniques: Team orienteering formulations [16] will be employed.
- Predictive Context: Unlike prior itinerary planners, this module integrates crowd, weather, and price forecasts to ensure itineraries are not only feasible but optimized under likely future conditions [17].
- Output: Sequenced daily itineraries (e.g., morning–afternoon–evening) balancing efficiency, satisfaction, and sustainability.
- This represents a methodological novelty: itineraries are adaptive and foresight-driven rather than static.

5. **Sustainability Assessment & Overtourism Mitigation Module**

- This module calculates a sustainability score for each recommendation or itinerary component, penalizing overcrowded sites and rewarding eco-friendly or community-based options [18][9]. It supports:
- Overtourism Mitigation Engine – reschedules or suggests alternatives when thresholds are exceeded.
- Local Benefit Optimizer – boosts local community-driven businesses and homestays.

6. Micro-Moment Context and Proactivity Handler

- This layer listens for contextual triggers ("It's lunchtime near Galle," "Rain expected in 15 minutes") and proactively queries the optimization and itinerary modules for quick, context-aware suggestions. Notification delivery respects user preferences to avoid overload [3][4].

7. Recommendation Delivery and Interface Layer

- Final recommendations and itineraries are delivered via a multilingual, multimodal interface (mobile/web app or chatbot). Features include:
- Ranked lists, interactive maps, and daily itineraries.
- Natural language explanations (e.g., "Recommended because it matches your hiking interest and is less crowded this afternoon [9]").
- Knowledge graph integration for semantic consistency and explainability [1].
- Exploration of neuro-symbolic explainability to combine reasoning with adaptability [19].
- Feedback collection for iterative learning.

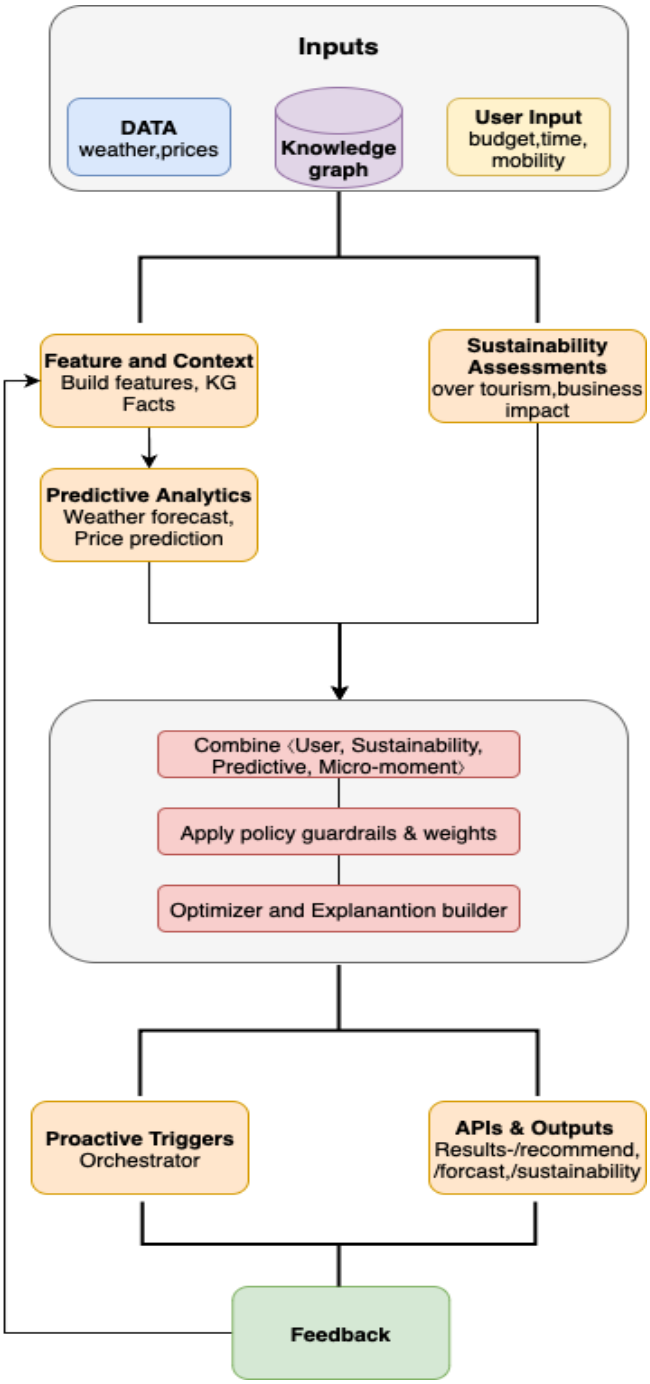

Figure 3.2: Component Diagram

Technology Stack Considerations

- Programming & Frameworks: Python (FastAPI, PyTorch, PyMoo), React Native/web front-end.
- Data Handling: Kafka + Flink for streams, InfluxDB for temporal data, PostgreSQL for profiles, Neo4j for tourism knowledge graph.

- Optimization Libraries: DEAP and PyMoo for GA and NSGA-II implementations.
- Deployment: Docker + Kubernetes for scalability.
- Caching & Fast Queries: Redis/MongoDB.
- Feedback Loop & Learning
- User interactions (implicit GPS traces, explicit likes/dislikes) refine collaborative filtering and adjust recommendation weights. Over time, reinforcement learning can tune the objective weights, further improving itineraries and proactive suggestions.

Methodological Novelty

- Predictive Itinerary Planning: Integrating forecasts into multi-day tour planning [16][17].
- Sustainability + Knowledge Graph: Embedding responsible tourism rules and semantic explainability [1].
- Neuro-Symbolic Proactivity: Combining reasoning and learning for adaptive, context-aware recommendations [19].
- Micro-Moment Engagement: Delivering timely, proactive recommendations in real time [3][4].

Development Plan & Experiments

- Phase 1: Data ingestion + predictive models.
- Phase 2: Baseline recommender (multi-objective).
- Phase 3: Add sustainability + itinerary planning
- Phase 4: Proactive micro-moment module.
- Phase 5: User interface + explainability.
- Phase 6: Evaluation via offline experiments and pilot studies.

4.PROJECT REQUIREMENTS

In developing the proposed solution, we have identified a comprehensive set of requirements that the system must fulfill. These include user requirements (capturing the needs and expectations of travelers and other stakeholders) and system requirements (defining the technical capabilities and constraints necessary to support those needs). Within these categories, the requirements are further detailed as functional requirements (features and behaviors of the system) and non-functional requirements (system qualities and performance constraints). To support the requirements definition, design artifacts such as use case scenarios, test cases, and tentative wireframes have been prepared. Typical user personas were also considered to ensure that the solution addresses diverse traveler needs. All requirements were derived from the research objectives and the intended deployment context of a conversational tourism assistant.

4.1 Functional Requirements

The functional requirements describe what the system should do:

- **FR1: User Profile Management** – Create and maintain user profiles, including preferences, budget, language, and accessibility needs.
- **FR2: Real-Time Data Acquisition** – Continuously fetch and update weather, events, crowd levels, and pricing data. Real-time streaming has been shown to be critical for foresight-driven recommendation systems [17].
- **FR3: Recommendation Generation** – Generate personalized lists of attractions, restaurants, and activities using the multi-criteria optimization algorithm.
- **FR4: Multi-Modal Query Handling** – Accept queries via text, voice, and images (e.g., photo-based matching of attractions).
- **FR5: Contextual Proactive Suggestions** – Push relevant suggestions based on time, location, and events (e.g., dining, rain alerts, cultural shows) [3][4].
- **FR6: Itinerary Planning** – Build multi-day, optimized itineraries on request. The planner must sequence visits (morning–afternoon–evening), consider travel times,

opening hours, and user constraints (budget, duration), and adapt dynamically with predictive context (weather, crowd, price forecasts) [16][17].

- **FR7: Sustainability Mode** – Provide eco-friendly and community-driven options with a "green travel" toggle, building on sustainability-aware tourism recommendation practices [18][9].
- **FR8: Explanation and Justification** – Provide rationale for each recommendation (e.g., "Suggested because it avoids peak crowd hours"), aligning with explainable recommendation practices [1][18].
- **FR9: Feedback Collection** – Collect explicit (ratings, likes/dislikes) and implicit (clicks, ignores) feedback for continuous improvement.
- **FR10: Security and Privacy Compliance** – Ensure user data protection and compliance with privacy norms.
- **FR11: System Admin Functions** – Tools to update the knowledge graph, monitor system health, and adjust weights/rules [1].

4.2 Non-Functional Requirements

The non-functional requirements specify qualities and constraints for the system:

- **NFR1: Performance** – Recommendations within 2 seconds; 95th percentile under 5s even at peak.
- **NFR2: Availability and Reliability** – High availability with graceful degradation (fallback strategies).
- **NFR3: Accuracy and Relevance** – forecasting targets defined in objectives. Predictive accuracy is central to foresight-driven systems [17].
- **NFR4: Usability and UX** – Conversational interface must be simple, multilingual, and intuitive. Proactive alerts must be helpful but not intrusive [3][4].
- **NFR5: Security** – HTTPS, secure authentication, and protection against vulnerabilities.

- **NFR7: Maintainability and Extensibility** – Modular design, easy integration of new data sources, and documented schemas.
- **NFR8: Commercial Viability** – Architecture should support future monetization via partnerships and affiliate integrations.

4.3 System Requirements

- Support multilingual natural language processing for Sinhala, Tamil, and English.
- Continuously ingest and process real-time tourism data (weather, crowds, prices, events).
- Generate recommendations using predictive analytics and multi-objective optimization.
- Provide itinerary planning that integrates predictive forecasts (weather, price, crowd).
- Ensure high performance (≤2s responses), secure data handling, and scalability for concurrent users.

4.4 User Requirements

- Traveler (general user): Must be able to interact with the system in Sinhala, Tamil, or English via text or voice.
- Traveler (planner): Must be able to generate personalized, multi-day itineraries that respect budget, preferences, and trip duration.
- Traveler (on-trip): Must receive proactive, context-aware suggestions triggered by weather, location, or time.
- Eco-conscious traveler: Must see sustainability indicators and eco-friendly alternatives when available.
- All users: Must receive clear explanations for recommendations and be able to give feedback to refine results.

5. GANTT CHART

Task Name	Progress	Mar	Apr	May	Jun	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
Project Initiation																	
Study about research gaps	100%																
Deciding appropriate research area	100%																
Going through research papers	100%																
Finalizing a topic	100%																
Requirement Gathering & Analysis																	
Identifying functional requirements	100%																
Identifying non-functional requirements	100%																
Identifying data requirements	100%																
Identifying knowledge requirements	100%																
Design																	
Creating the Figma prototype	0%																
Development																	
Recommendation Service	0%																
Backend Development	0%																
Frontend Development	0%																
Testing																	
Unit and API testing	0%																
Integrating Testing	0%																
Load testing	0%																
User Acceptance Testing	0%																
Deployment	0%																

Figure 5.1: Gantt Chart

5.1 Work Breakdown Structure

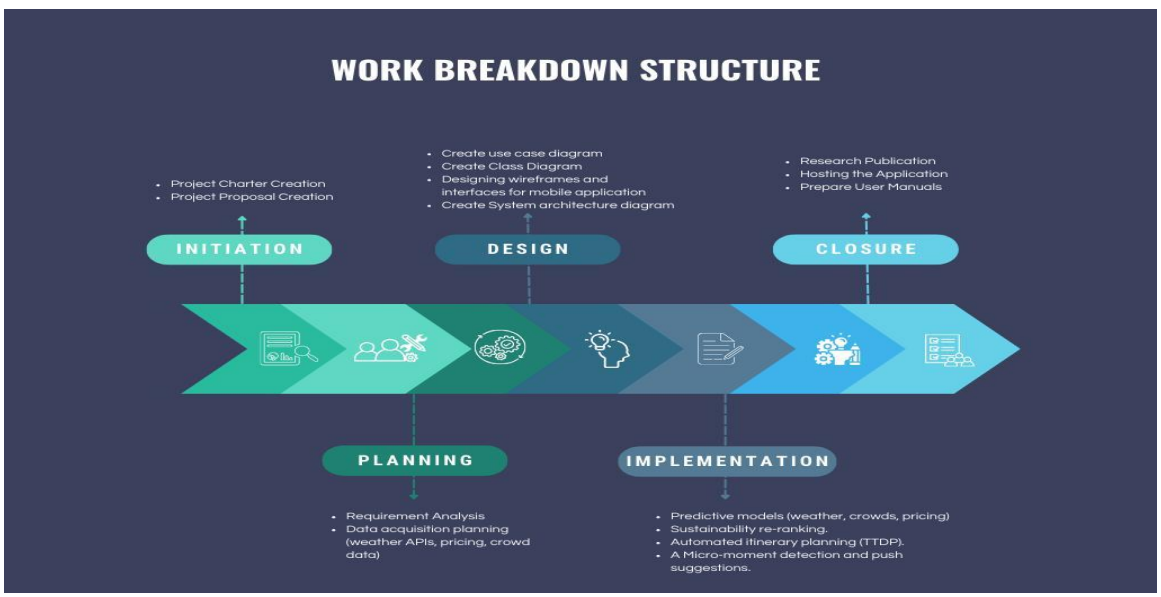

Figure 5.2 Work breakdown structure

6.PERSONAL AND FACILITIES

Facilitators:

- Dr. Lakmini Abeywardhena – Sri Lanka institute of Information Technology (SLIIT)
- Dr. Nathali Silva - Sri Lanka institute of Information Technology (SLIIT)

7. BUDGET AND BUDGET JUSTIFICATION

The proposed system has been planned with a lean but realistic budget, making use of free tiers where possible, while allocating essential costs for cloud services, APIs, and mobile deployment. The emphasis is on using **Azure cloud services** for backend hosting, storage, and speech capabilities, combined with **OpenAI API keys** for multi-agent reasoning and natural language tasks. In addition, a **Flutter-based mobile application** will be developed as the final deliverable, requiring costs for developer accounts on app stores. Hardware requirements such as microphones are also included to support data collection and testing. A buffer allocation has been added to account for unforeseen expenses during integration and field validation.

Description	Estimated Cost	Notes
Azure Cloud Services (VM hosting, Blob Storage, App Services, Speech-to-Text)	LKR 3000 / monthly (with free tier credits initially)	Backend hosting, data storage, STT/OCR services

OpenAI API Keys	LKR 3000	For embeddings, orchestration, reasoning tasks
Microphones for Voice Data Collection	LKR 10,000	For Sinhala–English code-mixed dataset collection
Mobile Testing Device	LKR 20,000	For app testing and evaluation
Miscellaneous / Buffer	LKR 10,000	Documentation, SD cards, adapters, printing

Table 7.1Budget Table

Total Budget = LKR 46,000

Budget Justification

Azure Cloud Services: Azure has been selected as the primary cloud provider due to its generous free tiers and reliable integration of storage, compute, and AI services. Virtual Machines and App Services will host the backend and APIs, while Blob Storage will hold datasets and graph exports. Azure Speech-to-Text will support Sinhala–English transcription at competitive rates, making it essential for the multilingual component.

OpenAI API Keys: A developer-tier plan will be used for embedding generation, agent orchestration, and conversational reasoning tasks. This ensures that the system can perform advanced semantic queries and multi-agent coordination effectively.

Flutter Mobile Application: The final deliverable will be a cross-platform mobile application built using Flutter, allowing deployment on both Android and iOS. Publishing costs include the Apple Developer Program ($99/year) and Google Play Developer registration ($25 one-time). These are unavoidable costs to ensure the app can be distributed officially and maintained with regular updates.

Microphones & Mobile Device: A high-quality microphone (~LKR 10,000) is required for collecting Sinhala–English speech data to fine-tune STT components. Additionally, a mid-range Android device (~LKR 20,000) is budgeted for testing the mobile application in real-world conditions.

Miscellaneous Costs: A buffer of $100 is included for documentation, printing, SD cards, adapters, and other unforeseen requirements during development and evaluation.

7. REFERENCES

[1] Lei, X., & Kamsin, A. Architecting Next-Generation Tourism Recommendation Systems: A Knowledge Graph Approach.

[2 Li, C., & Zheng, W. (2025). Nipping trouble in the bud: A proactive tourism recommender system. *Information & Management*, *62*(1), 104062.

[3] Lin, Y. L., & Lee, S. W. (2023). A personalized interaction mechanism framework for micro-moment recommender systems. *ACM Transactions on Interactive Intelligent Systems*, *13*(1), 1-28.

[4] IxDF, "What are Micro-moments? — updated 2025," *Interaction Design Foundation*, 2025. [Online]. Available: https://www.interaction-design.org/literature/topics/micro-moments

[5] Guizzardi, A., Pons, F. M. E., Angelini, G., & Ranieri, E. (2021). Big data from dynamic pricing: A smart approach to tourism demand forecasting. *International Journal of Forecasting*, *37*(3), 1049-1060.

[6] Gavalas, D., Konstantopoulos, C., Mastakas, K., & Pantziou, G. (2014). Mobile recommender systems in tourism. *Journal of network and computer applications*, *39*, 319-333.

[7] Sun, S., Li, Y., Guo, J. E., & Wang, S. (2022). Tourism demand forecasting: An ensemble deep learning approach. *Tourism Economics*, *28*(8), 2021-2049.

[8] Juan, Z., Zhang, J., & Gao, M. (2024). A multimodal travel route recommendation system leveraging visual Transformers and self-attention mechanisms. *Frontiers in Neurorobotics*, *18*, 1439195.

[9] Dalla Vecchia, A., Migliorini, S., Quintarelli, E., Gambini, M., & Belussi, A. (2024). Promoting sustainable tourism by recommending sequences of attractions with deep reinforcement learning. *Information Technology & Tourism*, *26*(3), 449-484.

[10] Cui, Y., Liang, S., & Zhang, Y. (2024). Multimodal representation learning for tourism recommendation with two-tower architecture. *Plos one*, *19*(2), e0299370.

[11] Cui, Y., Liang, S., & Zhang, Y. (2024). Multimodal representation learning for tourism recommendation with two-tower architecture. *Plos one*, *19*(2), e0299370.

[12] Yochum, P., Chang, L., Gu, T., & Zhu, M. (2020). Linked open data in location-based recommendation system on tourism domain: A survey. *IEEE Access*, *8*, 16409-16439.

[13] Permana, K. E., Rahmat, A. B., Wicaksana, D. A., & Ardianto, D. (2024). Collaborative filtering-based Madura Island tourism recommendation system using RecommenderNet. In *BIO Web of Conferences* (Vol. 146, p. 01080). EDP Sciences.

[14] Abbasi-Moud, Z., Vahdat-Nejad, H., & Sadri, J. (2021). Tourism recommendation system based on semantic clustering and sentiment analysis. *Expert Systems with Applications*, *167*, 114324.

[15] Guizzardi, A., Pons, F. M. E., Angelini, G., & Ranieri, E. (2021). Big data from dynamic pricing: A smart approach to tourism demand forecasting. *International Journal of Forecasting*, *37*(3), 1049-1060.

[16] Marzal, E., & Sebastia, L. (2024). Solving the tourist trip design problem with time windows and variable profit using incremental local search. *Applied Soft Computing*, *155*, 111399.

[17] Yoon, J., & Choi, C. (2023). Real-time context-aware recommendation system for tourism. *Sensors*, *23*(7), 3679.

[18] Nguyen, L. V. (2024). OurSCARA: Awareness-Based Recommendation Services for Sustainable Tourism. *World*, *5*(2), 471-482.

[19] Bhuyan, B. P., Ramdane-Cherif, A., Tomar, R., & Singh, T. P. (2024). Neuro-symbolic artificial intelligence: a survey. *Neural Computing and Applications*, *36*(21), 12809-12844.